Question	Answer	Marks
6(a)	$F_B = F_E$	**B1**
	either: $Bqu = qE$ and $E = V / d$ leading to $u = V / Bd$ *or*: $Bqu = qV / d$ leading to $u = V / Bd$	**B1**

Question	Answer	Marks
6(b)	$E_K = ½mu^2$	C1
	$u = \sqrt{[(2 \times 4.1 \times 10^{-17}) / (3.2 \times 10^{-27})]}$ $= 1.6 \times 10^5 \,ms^{-1}$	C1
	$B = 980 / (3.6 \times 10^{-2} \times 1.6 \times 10^5)$ $= 0.17\,T$	A1
6(c)	expression is independent of mass and charge	A1
6(d)	*either*: electric <u>force</u> is downwards so magnetic force is upwards *or*: no resultant <u>force</u> so magnetic force is upwards	B1
	(positive ions so) current is from left to right	B1
	from (Fleming's) left-hand rule, magnetic field is into the page	B1
6(e)	curved path inside plates with consistent direction of curvature and with no discontinuity at entry or in curvature	B1
	direction of deflection is upwards	B1

Question	Answer	Marks